

GAA Custom Electronics, LLC

Resources & Capabilities Overview

101904 Wiser Pkwy, Suites 105 & 106 • Kennewick, WA 99338 • Established 2014

Company Overview

GAA Custom Electronics, LLC is a specialized engineering firm founded in 2014, located in Kennewick, Washington. We serve the mass spectrometry research community — including universities, national laboratories, and instrument manufacturers — with high-performance custom electronics that accelerate R&D programs and enable entirely new experimental capabilities. Our compact, expert team combines deep electrical engineering expertise with hands-on fabrication resources, allowing us to take a project from concept to working prototype with exceptional speed and precision.

Our flagship product line, the Modular Intelligent Power Supply (MIPS) system, is in use at research institutions worldwide, with over 200 systems deployed and hundreds of man-years of cumulative field runtime. Beyond MIPS, we develop a full range of electronic subsystems — RF generators, FET-based pulse generators, electrometers, SLIM reversers, and more — each engineered to interface seamlessly with mass spectrometers and ion mobility instruments.

We are committed to open-source principles: hardware schematics, firmware, and host software are maintained publicly on GitHub (github.com/GordonAnderson), and our competitive pricing reflects our lean overhead model. First consultations — up to 30 minutes — are always free.

Engineering Capabilities

Electronic Design

Our electronic design capability spans the full signal chain, from precision analog front ends to complex digital logic and high-voltage power electronics. We have extensive experience designing circuits for demanding scientific instrumentation environments where noise, timing accuracy, and reliability are paramount.

Analog Circuit Design	Digital Circuit Design
PLC (Programmable Logic Controller) Systems	PLD / FPGA Design and Implementation
SPICE Circuit Simulation & Verification	Printed Circuit Board Design (Schematic to Layout)
High-Voltage Power Supply Design	FET-Based Pulse Generator Design (10–15 ns rise times)

RF Generator Design (ion mobility / quad transfer)	Prototyping and Low-Volume PCB Fabrication
Quadrupole Mass Filter Electronics	Packaging, Panel Machining & Assembly

Software Development

Our software capability covers the full stack from bare-metal firmware on microcontrollers to rich host applications with scripting, custom GUI panels, and multi-instrument coordination. We develop real-time embedded control code and the host applications that communicate with it — providing a complete, integrated solution.

Domain	Languages & Platforms	Notes
Embedded Firmware	C, C++, Assembly	ARM, ATMEL, Microchip microcontrollers
Real-Time Control	C, C++	Deterministic timing for ion optics control
Host Application	Qt (C++)	MIPS host GUI with scripting engine
Scripting & Automation	JavaScript	Custom panels, macro automation, UI scripting
Scientific Scripting	Python	Data acquisition, analysis, automation
Mobile / Wireless	iOS Objective-C	Wireless MIPS control interfaces
Windows Integration	Microsoft .NET, VBA	Excel integration, COM automation
PCB & Firmware Tools	Atmel Studio, MPLAB, Arduino IDE	Full toolchain for MCU development

Mechanical Design & Fabrication

In addition to electronics, we design and fabricate custom mechanical hardware including instrument panels, enclosures, vacuum flanges, and custom fixtures. Our in-house 3D CAD capability ensures that electronics and mechanical designs are co-developed for optimal fit and function.

3D CAD Design (parametric solid modeling)	Custom Instrument Panel & Enclosure Design
FDM Additive 3D Printing	Resin / SLA 3D Printing
Precision Milled Parts (CNC and manual)	Vacuum Feedthrough & Flange Fabrication

Connectivity & Communications

All MIPS-based systems support multiple host communication interfaces. Instruments can be networked together and controlled from a single unified host application.

- USB (primary wired interface)
- Wi-Fi (802.11 wireless)
- Ethernet (TCP/IP, suitable for lab networks and remote operation)
- Multi-instrument coordination via a single MIPS host application session

Facility & Resources

GAA Custom Electronics operates a 1,800 square foot, fully equipped instrument development laboratory — comprising two attached units (Suites 105 & 106) — in Kennewick, Washington. The facility is purpose-built for rapid electronics development and low-volume production, and is augmented by established relationships with local contract manufacturers for higher-volume requirements. Every stage of development — design, simulation, fabrication, assembly, and test — can be completed in-house.



Electronics R&D laboratory — oscilloscopes, spectrum analyzer, logic analyzer, and full bench test capability

Electronics Test Equipment

A well-stocked bench allows thorough characterization of analog, digital, RF, and power electronics:

Oscilloscopes (digital storage)	Spectrum Analyzer
Logic Analyzer	Arbitrary Waveform / Function Generators
Pulse Generators	Bench Power Supplies (multiple channels)
Precision Soldering & De-soldering Stations	Hot-Air Rework Stations

LCR Meter	Digital Multimeters
-----------	---------------------

Design & Development Tools

- Eagle / Cadence Allegro / KiCad — Schematic capture and multi-layer PCB layout
- SPICE simulation software for analog and mixed-signal verification
- 3D CAD software for mechanical and enclosure design
- Atmel Studio IDE (AVR / ARM microcontrollers)
- MPLAB X IDE (Microchip PIC / dsPIC families)
- Arduino IDE (rapid prototyping and open-source hardware)
- Qt Creator — host application development
- GitHub — version control and open-source publication (github.com/GordonAnderson)

Fabrication Resources

Complete in-house fabrication capability from PCB assembly to mechanical parts:

Equipment	Capability
Automatic Pick-and-Place Robot	SMT component placement for PCB assembly
Manual Pick-and-Place Station	Hand placement for fine-pitch and odd-form components
Reflow Solder Oven	Controlled-profile SMT soldering
3-Axis CNC Mill Table	Precision panel cutouts, mechanical parts, fixtures
Knee Mill with Digital Read-Out (DRO)	Manual milling for prototypes and one-off parts
Lathe with Digital Read-Out (DRO)	Turned parts, shafts, custom fittings
Drill Press	PCB drilling, panel work
FDM 3D Printers	Rapid plastic prototyping, brackets, enclosures
Resin (SLA) 3D Printer	High-resolution parts, fine-detail prototypes
PCB Fabrication	Boards ordered via PCBWay; typically 5–10 day turnaround



HW-T4-50F automatic pick-and-place robot for SMT assembly



Mill and lathe with digital read-out — machine shop

Mass Spectrometry Development Instruments

Access to research-grade mass spectrometers in-house allows us to develop and validate our electronics in the exact application environment our customers use. This is a rare and significant advantage for instrument electronics development.

- SCIEX 6500 Triple Quadrupole (high-sensitivity quantitation platform)
- SCIEX 6600 Triple TOF (high-resolution time-of-flight platform)
- Thermo MSQ Quadrupole × 3 (instrument development and electronics testing)



In-house SCIEX 6500 triple quad and 6600 Triple TOF — a rare and decisive advantage for electronics development and validation

Vacuum System Development Resources

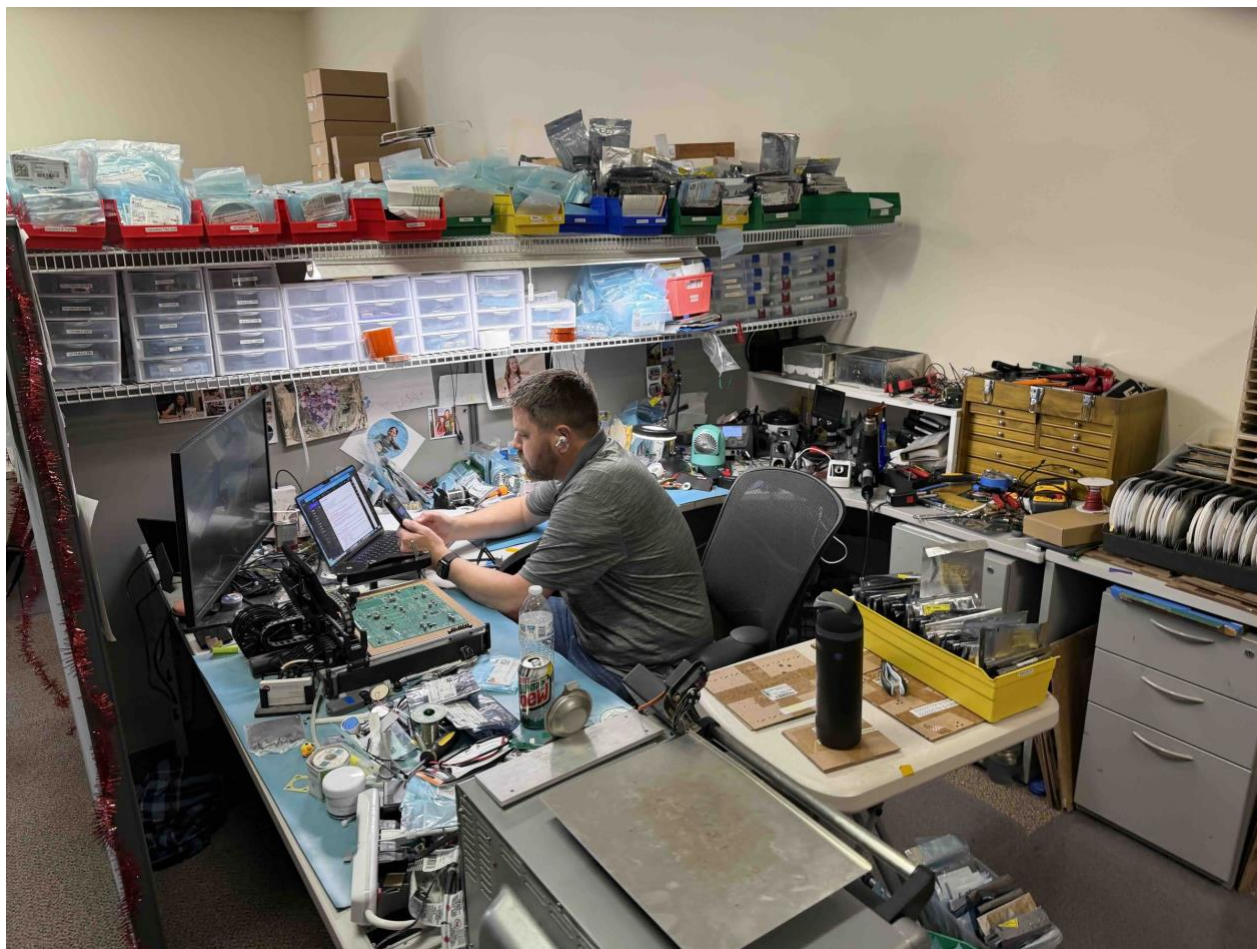
Many mass spectrometer electronics projects require vacuum-compatible hardware and testing. Our vacuum development resources allow in-house testing of ion optics, feedthroughs, and high-voltage electronics under realistic operating pressures.

Mechanical Vacuum Pumps (roughing)	Turbomolecular Pumps (high vacuum)
Custom Vacuum Test Chambers	Ion Gauge & Thermocouple Vacuum Gauges
Electrical Feedthroughs (various types)	High-Voltage Vacuum-Compatible Assemblies

Quality, Support & Delivery

GAA Custom Electronics is committed to delivering reliable, well-documented electronics backed by responsive support. Our quality practices and support commitments include:

- One-year warranty on all products covering materials and workmanship defects
- Free firmware bug fixes and phone support for the warranty period
- Comprehensive operations manuals provided with every product
- Example firmware, scripts, and documentation delivered on USB drives and via Google Drive
- Rapid iteration capability: prototype-to-test cycle measured in days, not weeks
- Open-source hardware and software — customers are never locked in
- Established relationships with local contract manufacturers for volume scale-up
- Competitive, transparent pricing — detailed rate sheet available on request



Active system assembly — organized component inventory and dedicated integration bench

Contact & Engagement

We welcome inquiries from researchers, engineers, and instrument manufacturers at any stage of a project.

Address	101904 Wiser Pkwy, Suites 105 & 106 Kennewick, WA 99338

Phone	509.628.6851
Email	gaa@gaa-ce.com
Website	www.GAAcustom.com
GitHub (open-source code)	github.com/GordonAnderson
Initial Consultation	First 30 minutes always free

Please visit our website or contact us directly to discuss your project requirements and receive a prompt, detailed quote.